SKILL WALLET
A SmartBridge Product

SMARTBRIDGE
Let's Bridge the Gap

Problem-Solution Fit

Date	30 june 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] ☐ Retail Bank Credit Underwriters and Risk Officers. ☐ Bank Operations Managers handling credit card enrollment divisions. ☐ Financial Institutions looking to transition from rigid legacy scoring to modern data-driven automation.	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** ☐ Technical Constraints: End-users are credit operations agents, not software engineers or data scientists. They cannot interact with terminal scripts or raw command-line tools. ☐ Environment Constraints: Limited processing power on standard office desktop computers, requiring a highly efficient application that runs without expensive server upgrades.	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** ☐ **Solution A: Manual Folder Verification Check** • *Pros:* High human oversight. • *Cons:* Exceptionally slow, expensive scale costs, prone to human error, and creates long customer wait times. ☐ **Solution B: Traditional Static Rule Engines (e.g., FICO cutoffs)** • *Pros:* Fast and cheap to compute.

- *Cons:* Overly rigid; cannot learn complex interactions between variables like income bracket, age, and employment duration.

2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]

- Pains: High application backlogs due to cross-referencing multi-table user profile charts line by line. Manual fatigue frequently leads to human error or approval biases.
- Frequency: Continuous. Occurs daily with every new application cycle, spiking significantly during seasonal credit card marketing promotions.

9. PROBLEM ROOT / CAUSE [RC]

- **Root Cause:** A complete disconnect between data storage formats. The primary demographic data sits in one place, while transactional payment logs sit in a separate, multi-row time-series table.
- There is no unified automation layer capable of instantly joining these tables, fixing missing records, and rendering an immediate risk classification.

7. BEHAVIOR
+ ITS INTENSITY **[BE]**

- **Behavior:** Agents manually open and cross-examine separate tables (application_record.csv and credit_record.csv) to map historical defaults.
- **Intensity:** High intensity. This repetitive, manual workflow consumes up to 30 minutes per applicant, leading to cognitive fatigue by mid-shift.

3. TRIGGERS TO ACT [TR]

- Surges in processing backlog times where customers have to wait days or weeks for credit card approval decisions.
- An increase in credit defaults (Non-Performing Assets) caused by poor manual risk identification or missed warning signs in repayment history logs.

4. EMOTIONS
BEFORE / AFTER **[EM]**

10. YOUR SOLUTION [SL]

- An end-to-end Machine Learning web platform. The Python engine automatically cleans, aggregates, and encodes the datasets, training a robust **Random Forest Classifier with balanced class weights** to catch high-risk behaviors.
- This logic is bundled into a lightweight file (model.pkl) and wrapped in an intuitive **Flask Web Dashboard**. Credit officers can type applicant profiles into a clean UI form and

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- **Online:** Accessing localized bank intranets, managing spreadsheet files, and navigating separate internal account lookup screens.

OFFLINE
Offline: Discussing complex or borderline applicant folders with senior

Before: Stressed, overwhelmed by high file volumes, and anxious about making calculation errors or missing delinquency markers. After: Confident, highly efficient, and empowered by an objective machine learning engine that delivers lightning-fast screening recommendations**.**	click "Predict" to receive an absolute, instantaneous evaluation verdict.	managers during risk assessment team syncs.